Infosys - Coding 5

HackerRank

A specialized vending machine is designed to dispense exactly T units of credit using a set of N available token denominations.

Each denomination i (from 1 to N) is characterized by a specific value d[i] and a limited quantity c[i] representing the number of tokens of that value currently stored in the machine.

The machine operates under a strict protocol: a token of value v can only be used if v is a divisor of the current remaining amount to be dispensed. Furthermore, the machine follows a rigid greedy selection process:

1. At any point, it identifies the largest denomination d[i] that satisfies the divisibility rule and has a non-zero quantity remaining (c[i] > 0).
2. It then dispenses as many tokens of this chosen denomination as possible — either until the denomination is exhausted (its quantity c[i] becomes zero) or until the total remaining amount to be dispensed reaches zero.

Your task is to determine the total number of tokens the machine will dispense to reach the target T exactly. If the machine reaches a state where the remaining amount is greater than zero but no available denomination divides it, the machine fails and you should return -1.

Find the total number of tokens dispensed, or -1 if impossible.

Input Format

First line: integer N, the number of available token denominations.

Second line: integer T, the target total credit units to dispense.

Next N lines: two space-separated integers per line — value and quantity — representing row i of denominations.

Constraints

1 <= N <= 100

1 <= T <= 10^9

1 <= denominations[i][j] <= 10^9

Output Format

The output contains an integer value

Sample Input 0

```
2
10
```

```
5 2
2 3
```

Sample Output 0

```
2
```

Explanation 0

Starting with a target of 10, the machine identifies 5 as the largest available denomination that divides 10. It then dispenses two tokens of value 5, fully satisfying the target and resulting in a total of 2 tokens.

Sample Input 1

```
3
20
10 1
5 5
2 10
```

Sample Output 1

```
3
```

Explanation 1

The machine first uses one token of value 10 to reduce the target from 20 to 10. With 10 remaining, it then selects 5 as the largest divisor and dispenses two tokens of value 5, reaching the target exactly with a total of 3 tokens.